Cutting the Cord: Authority-Independent, Escrow-Free Authentication and Key Management for TUAV-Assisted Infrastructure-Less IoV

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***Abstract*—A recent scheme by Tan, Zheng and Vijayakumar (IEEE T-ITS 2023) uses a tethered UAV (TUAV) to replace roadside units destroyed by a disaster, so that an Internet of Vehicles keeps working "without infrastructure". We show that it does not achieve this. Every authentication is relayed over a satellite link to a cloud Trusted Authority (TA) that computes all pairings. The UAV credential depends only on values the TA holds, so the TA can impersonate any UAV. The CRT-based group key reveals every member's decrypting key to every other member, so a revoked UAV still obtains the new key. The reported costs omit the TA's pairings and an $O(n^3)$-byte key broadcast. We confirm these findings in code and, for the escrow attack, with ProVerif. We then propose *Cutting the Cord*, which moves the trust root offline. The TUAV authenticates UAVs itself with escrow-free, pairing-free certificateless signatures in a sound batch with a bounded fallback. It delivers the group key per member under fresh ECDH keys, admits vehicles through authenticated UAV relays, and hands members between TUAVs with single-use tickets. Removing the authority creates a problem of its own: nobody online can overrule a forged packet or a captured edge node. We demonstrate this with 7 executed attacks on our own first design, including one forged packet that locks a UAV out for an epoch and a captured TUAV that reads every neighbouring cell's group key. We fix them and add a containment tier in which a quorum of TUAVs evicts a captured peer with the TA offline. On a 25-cell grid, one captured TUAV then exposes 4.0% of cells instead of 15.7%. Symbolic ProVerif models of the core protocols verify authentication, key secrecy, forward secrecy and revocation, including handover with a captured TUAV, and delimit the residual TA power to Girault level 2. On the same curve and library, the base scheme's TA-inclusive cost is about 65× ours, our scheme needs no satellite traffic, and it sends 1,823× less key traffic.**

***Index Terms*—Internet of Vehicles, tethered UAV, certificateless signatures, batch verification, group key management, node capture, revocation, ProVerif.**

I. Introduction

Roadside units (RSUs) authenticate vehicles in a conventional Internet of Vehicles (IoV). After an earthquake or flood they are among the first things to fail, exactly when rescue traffic needs secure coordination. Tan *et al.* [1] propose a *tethered UAV* (TUAV), powered by cable from an emergency vehicle, as a flying edge server for a swarm of UAVs, with certificateless batch authentication and a CRT-based group key. They report 151 ms to authenticate 120 UAVs and eleven satisfied security properties.

We examined the scheme equation by equation, implemented it on BN254, and modelled its credential in ProVerif. The architecture is sound; the protocol is not. The TUAV cannot verify anyone without a satellite round trip to the TA (so authentication stops when the link fails). The TA can impersonate every UAV. A revoked UAV keeps receiving group keys. The efficiency figures omit the dominant cost.

Fixing this by taking the authority out of the field is not free. A cloud TA, for all its faults, could overrule a false report and refuse a stolen device. With the TA offline, a captured TUAV and a forged packet face no such check, and a design that ignores this fails in new ways. Our own first design did.

Contributions. (1) Six findings against [1], four reproduced in code and one found automatically by ProVerif (Sec. IV). (2) *Cutting the Cord*: a framework with an offline trust root and four tiers. C1 is autonomous, escrow-free UAV authentication with secure rekeying. C2 admits vehicles through UAV relays. C3 is signature-bound, ticket-based handover between TUAVs. C4 evicts a captured TUAV by quorum (Sec. V). (3) A compromise analysis of authority-independent admission: 7 executed attacks on our first design, their fixes, and the rule behind them: with no authority online, any state change that outlives a window must rest on evidence a third party can check (Sec. VI-A). (4) Security arguments and eight ProVerif models with twenty-three queries, which pin down the residual TA power precisely (Sec. VI). (5) An implementation of both schemes on one curve and library, with nine experiments, including the blast radius of captured TUAVs and the measured cost of containing them (Sec. VII).

II. Background and the Base Scheme

Certificateless cryptography [2] splits a private key into a partial key issued by the authority and a secret value the authority never sees. In Girault's terms [3], the authority is then at *level 2*: it cannot learn a user's key, but it could issue a second, detectable key. At level 1 it knows users' keys. **Batch verification** needs random small exponents to be sound [4]. **Group key management** ranges from per-member (unicast) delivery to key trees (LKH) [6] and single-broadcast CRT "secure locks" [7].

The base scheme [1] has a TA, a TUAV and n UAVs. In IS1–IS5 the TA gives the TUAV (id_{tu}, s_{tu}) and each UAV (id_i, s_i), and the TUAV broadcasts $(ts_{tu}, ID_{tu}, R_{tu}, Q_{tu})$. In US1–US3 each UAV picks r_i and sends $(ts_2, ID_i, \xi_i, \upsilon_i, \Gamma_i)$, where $\xi_i = h_1(r_i)$ and

$$\Gamma_i = [s_i h_3(ID_i, ts_2, s_i) R_{tu} - h_1(s_i \xi_i)]P - \upsilon_i h_1(r_i) Q_{tu}. \quad (1)$$

In GA1–GA2 the TUAV uploads the requests to the TA, which computes $\sigma_i = h_1(s_i\xi_i)P$ and three pairings ψ_i, μ_i, δ_i. In GA3 the TUAV checks one product equation. In GD1–GD6 the TUAV publishes the coefficients of

$$\Lambda(x) = \kappa_{tu} \textstyle\sum_i \omega_i\varphi_i + \prod_i (x - \sigma_i), \tag{2}$$

with $\omega_i = \prod_{j\neq i} \sigma_j$ and $\varphi_i \equiv \omega_i^{-1} \bmod \sigma_i$. Each UAV recovers $\kappa_{tu} = \Lambda(\sigma_i) \bmod \sigma_i$. In DU1 the polynomial is rebuilt over the remaining members' *unchanged* σ_i.

III. Related Work

Drone-assisted vehicular security. Recent schemes pursue lightweight authentication for UAV-assisted vehicular networks [15], [16] and batch authentication with tree-based key agreement for UAV swarms [14]. Swarm group authentication has also been built on threshold secret sharing [13]. We found no published cryptanalysis of [1], and no scheme that combines an offline authority with containment of a captured edge node for this setting.

CRT group keys. Secure locks [7] deliver one key to many recipients through CRT, and Vijayakumar *et al.* [10] reduced the server's rekeying cost. Xia *et al.* [11] report that in [10] a former member may still obtain the group key and colluding members may recover another member's long-term key. Our finding F4 belongs to the same class of failure. It targets the polynomial construction of [1] and needs one insider and no collusion. Newer CRT-based protocols pack per-member ciphertexts into one broadcast integer [17], a different construction that F4 does not address.

Revocation without an authority. Chan *et al.* [12] revoke compromised sensor nodes by neighbours' votes. C4 applies the quorum idea to TUAVs, using certificateless signatures so that any member can check an eviction with the TA's public key alone.

IV. Analysis of the Base Scheme

F1: not infrastructure-less. GA3 uses only values computed by the TA in GA2, and the TA is reached by satellite. With link availability A, a round succeeds with probability A, and every round pays a satellite round trip.

F2: the TA impersonates every UAV. Since $\xi_i = h_1(r_i)$ is sent in clear, (1) depends only on s_i, which the TA stores, and on public values. The TA can therefore build a valid request for any UAV (level 1, so claims F4 and F10 of [1] fail). Verification also needs s_i, so no credential is non-repudiable. Our implementation accepts a TA-built request, and ProVerif finds the attack (Q16, Table II).

F3: no forward secrecy. σ_i is determined by s_i and the public ξ_i; a later leak of s_i opens every recorded group key.

F4: revocation fails.

Lemma 1. *For every member j: $\Lambda(x) - \Lambda(\sigma_j) = \prod_i (x - \sigma_i)$.*

Proof. $\Lambda(x) = C + \prod_i (x - \sigma_i)$ with constant C, and the product vanishes at σ_j. □

Any member therefore recovers *all* σ_i by integer root finding. Because DU1 reuses them, a revoked UAV computes the new key as $\Lambda^*(\sigma_j) \bmod \sigma_j$ for any staying j. Our code executes this attack. The σ_i are also points in GA2 but moduli in GD1, so the construction is not even well defined as printed.

F5: empty proof, unsound batch. Theorem 1 of [1] reduces forgery to the CDH instance $(s_{tu}P, (\sum \upsilon_i\xi_i)P)$, whose second exponent is public, so the instance is trivial. GA3 has no random weights, so $\Gamma_1 + \Delta, \Gamma_2 - \Delta$ pass while both fail individually (reproduced). GA4 has no failure branch, so one bad request rejects everyone.

F6: cost omits the TA. GA2 performs three pairings per UAV. With the base paper's own MIRACL constants, group authentication of 120 UAVs costs 3720 ms plus the satellite RTT, not 151 ms. The CRT coefficients make each acknowledgement $O(n^2)$ bits, sent to each of n UAVs (Sec. VII).

All six share one root cause: *the TA is inside every operation.*

V. Cutting the Cord

Model. A TA (offline after deployment); TUAVs $\mathcal{T}_a$ with public identities; UAVs $\mathcal{U}_i$ and vehicles $\mathcal{V}_j$ with pseudonyms. The attacker is Dolev–Yao on all links, plus Type-I (key-replacing outsider) and Type-II (malicious TA) certificateless adversaries, insiders, and batch poisoners.

Setup and registration (offline). The TA holds s ($P_{\text{pub}} = sP$) to certify keys and t ($T_{\text{pub}} = tP$) to trace pseudonyms. For each epoch ep an entity picks x, sends $X = xP$ with a proof of possession, and receives $PID = (\rho P, RID \oplus H_0(\rho T_{\text{pub}} \,\|\, ep), role)$, $Y = yP$ and $d = y + sh$, where $h = H_1(PID, ep, X, Y)$. The full key is $sk = x + d$ and $pk = X + Y + hP_{\text{pub}} = skP$. The TA knows d but never x. Epochs start at 1; TUAV credentials use $ep = 0$, so the two kinds cannot stand in for each other.

C1, UAV tier. The TUAV broadcasts a signed beacon carrying a fresh $E_a = e_aP$. A UAV signs $m_i = (\texttt{join} \,\|\, ID_a \,\|\, E_a \,\|\, E_i)$ with a Schnorr-type signature (R_i, v_i), $v_i = r_i + c_i sk_i$ for challenge c_i, and sends $\langle PID_i, ep, X_i, Y_i, E_i, ts_i, R_i, v_i\rangle$, signing at most once per beacon. The TUAV filters the requests (freshness, epoch, eviction list, role), keeps at most $c = 2$ distinct requests per pseudonym, sampled uniformly from those received, verifies contested pseudonyms individually, and checks the rest, with random 64-bit λ_i,

$$\Big(\sum \lambda_i v_i\Big)P = \sum \lambda_i R_i + \sum (\lambda_i c_i)(X_i + Y_i) + \Big(\sum \lambda_i c_i h_i\Big)P_{\text{pub}}. \tag{3}$$

If the check fails, the TUAV verifies each request, admits the valid ones, and stays in individual mode until windows are clean. A failed signature evicts nobody (Sec. VI-A). The TUAV then derives $K_i = \mathsf{KDF}(e_aE_i \,\|\, \dots)$ and broadcasts one ACK with $\mathsf{Enc}(K_i, GK_a \,\|\, \nu)$ for each member. Whenever members join or leave, at most once per window, rekeying draws a fresh GK'_a and encrypts it only under the K_i of those who should hold it; members accept only increasing ν. LKH [6] reduces a single leave to $2\lceil \log_2 n \rceil - 1$ ciphertexts.

C2, vehicle tier. Vehicles hold the same kind of credential, and the certified role stops them from claiming the UAV tier. A UAV relays their signed requests in a bundle authenticated by $\mathsf{MAC}(K_i, \cdot)$, so the TUAV discards bundles from non-members before any signature work. Valid vehicles receive a separate regional key VK_a. Routine V2V beacons carry a MAC under VK_a; safety-critical events are signed.

C3, handover. Neighbouring TUAVs agree a link key K_{ab} by signed ECDH. Before a member $\mathcal{M}$ leaves, $\mathcal{T}_a$ gives it a ticket $\mathsf{Enc}(K_{ab}, PID \,\|\, pk \,\|\, hk \,\|\, N \,\|\, exp)$, where $hk = \mathsf{KDF}(K_{\mathcal{M}} \,\|\, \dots)$. With a fresh E, $\mathcal{M}$ signs $(ID_a \,\|\, ID_b \,\|\, H(\text{ticket}) \,\|\, E)$. $\mathcal{T}_b$ opens the ticket, rejects reused N, verifies a MAC under hk and that signature, runs one fresh ECDH, rekeys its cell, and sends the new group key with its own signature over $(ID_a \,\|\, ID_b \,\|\, H(\text{ticket}) \,\|\, E \,\|\, E_b)$, which $\mathcal{M}$ checks under ID_b. $\mathcal{T}_a$ rekeys once $\mathcal{M}$ has left. $\mathcal{T}_b$ never learns $K_{\mathcal{M}}$. The ticket spares the member the wait for a beacon; it no longer vouches for either side on its own.

C4, containment. A TUAV that learns of a peer's capture signs $(\texttt{evict-tuav} \,\|\, ID_x)$ under its $ep = 0$ credential. How it learns (tamper sensor, lost tether, crew report) is outside the protocol. Votes from t distinct other TUAVs form a certificate that anyone checks with P_{pub} alone, as a batch. TUAVs holding it drop K_{ax} and refuse $\mathcal{T}_x$'s tickets; members holding it refuse $\mathcal{T}_x$'s beacons.

Tracing and revocation. Offline, the TA recovers $RID = PID^2 \oplus H_0(tPID^1 \,\|\, ep)$ and can sign a list of the remaining pseudonyms of a revoked entity. A TUAV may evict locally on an operator's decision, but neighbours accept an eviction only with a TA-signed list or with evidence: two valid signatures by one pseudonym under one beacon, which an honest member never produces. Nothing waits for the TA.

VI. Security Analysis

Proposition 1. *No step of C1–C4 sends to or waits for the TA; verification needs only P_{pub}.*

Theorem 1 (Unforgeability)**.** *Under ECDLP in the ROM, no Type-I or Type-II adversary forges a signature under an honest entity's registered key.*

Sketch. Type II: embed the challenge in X^*; forking on H_2 yields sk^*, hence $x^* = sk^* - d^*$. Type I: embed it in P_{pub}; double forking on H_2 and on H_1 yields $(sk^* - sk^{*\prime})/(h^* - h^{*\prime})$ [8]. Unlike [1], the challenge is never a public value. □

Theorem 2 (Batch soundness [4])**.** *If any request is invalid, (3) holds with probability at most 2^{-63} (the weights are odd, leaving 63 free bits).*

Theorem 3 (Keys)**.** *Under CDH and AEAD security, K_i is known only to $\mathcal{T}_a$ and $\mathcal{U}_i$. Session keys survive later compromise of all long-term keys. A removed member cannot obtain GK'_a, and a joiner cannot obtain earlier keys.*

Proposition 2 (Bounded poisoning)**.** *Over n pseudonyms, a window costs at most $T_{BV}(n) + cnT_{SV}$, and a failed signature changes no state beyond the window. A pseudonym that no forged request names is always admitted. One named by k forged requests is admitted with probability $\min(1, c/(k+1))$ per window.*

Sketch. Reservoir sampling keeps each of the $k+1$ requests with equal probability, and a kept valid request is verified individually. □

Proposition 3 (Containment)**.** *Under Theorem 1, a set $\mathcal{C}$ of captured TUAVs holding no member credential obtains no group key of a cell outside $\mathcal{C}$. It cannot replace the pairwise key of a registered member at an honest TUAV, cannot add an honest member to an honest TUAV's eviction list, and, if $|\mathcal{C}| < t$, cannot evict an honest TUAV.*

Sketch. Each outcome needs a forgery. Handover admission needs a fresh signature bound to the ticket, the destination and E, under a key the TA certified at $ep \geq 1$; the TUAV's own key is certified at $ep = 0$. Evidence needs two signatures under one beacon, and an honest member signs once. An eviction needs t distinct $ep = 0$ signatures. □

A. Pitfalls of removing the authority

Reviewing our first design adversarially, we found the attacks in Table I. All were executed against its implementation. A1–A4 are design flaws. A5 and A6 are gaps between its specification and its code, which its tests did not cover. A7 survived the fix for A2 and was found by ProVerif: the member authenticated the new TUAV only through hk, which the captured old TUAV also derives.

A1 deserves attention because the design looked safe. It evicted a pseudonym whose signature failed, as a defence against poisoning. But pseudonyms travel in the clear and stay valid for a whole epoch, and a failed signature proves nothing about the pseudonym it names. Any outsider who has heard a UAV once could evict it with one packet, and eviction sharing spread the damage to neighbouring cells. A2 and A3 follow the same pattern. A ticket let $\mathcal{T}_b$ skip the member's signature, so whoever held K_{ab} could admit anyone, and neighbours took eviction lists on trust. We had described a captured TUAV as exposing its own cell; through the link keys it exposed every neighbour.

The common rule: *with no authority online, any state change that outlives a window must rest on evidence a third party can check.* A failed signature is not such evidence, and neither is a neighbour's word.

ProVerif [9]**.** Table II lists the twenty-three queries. Q7 and Q9 fail *as predicted*. A malicious TA can still issue a second key for a pseudonym, or enrol a phantom member and receive the group key. This is inherent to certificateless schemes and detectable. Q6 and Q8 show that it cannot sign under the registered key or read pairwise traffic. Q12 holds only non-injectively, because ProVerif's table lookup and insert are not atomic. Our TUAV processes handovers sequentially, and a replayed ticket still yields no key (Q13–Q15). Q12–Q15 model tickets with K_{ab} secret and without the member's signature, so they could not reveal A2, which needs a captured $\mathcal{T}_a$.

TABLE I

ATTACKS ON OUR FIRST DESIGN. A1–A6 ARE EXECUTED IN E8A, A7 IN THE TESTS; ALL SUCCEED THERE AND FAIL AFTER THE FIX.

#	Attack	Fix
A1	One forged request naming a victim evicts it for the epoch; eviction spreads	A failure evicts nobody; sample c per pseudonym
A2	Captured TUAV mints tickets: reads a neighbour's group key, replaces a member's pairwise key	Member signs each handover
A3	Captured TUAV sends a false eviction list	Merge only evidence or TA-signed lists
A4	Vehicle credential claims the UAV role and gets GK_a	Certify the role in PID
A5	Replayed rekey restores a key a removed member holds	Versions only increase
A6	Joiner receives the key already in use	Rekey on arrival, once per window
A7	Captured TUAV answers its departing member in the neighbour's name and reads its traffic (found by ProVerif)	New TUAV signs its reply

TABLE II

PROVERIF 2.05 RESULTS. Q7 AND Q9 FAIL AS PREDICTED; Q12 IS WEAKER THAN THE INJECTIVE AGREEMENT WE WANTED; Q17–Q18 ARE ATTACKS ON THE FIRST DESIGN'S C3.

#	Property	Result
Q1–Q3	C1 mutual authentication (Type I, injective)	✓ true
Q4–Q5	group-key secrecy; forward secrecy (all keys leaked)	✓ true
Q6	malicious TA cannot use the *registered* key	✓ true
Q7	malicious TA with a *new* key (key replacement)	false (level 2)
Q8	pairwise-key data secret from the TA	✓ true
Q9	group key vs. phantom member enrolled by the TA	false (level 2)
Q10–Q11	revoked member: old key known, new key secret	✓ true
Q12	handover acceptance ⇒ member started it	✓ non-inj.
Q13–Q15	handover completion (inj.), secrecy, forward secrecy	✓ true
Q16	**base scheme**: acceptance ⇒ UAV sent	✗ attack
C3 with $\mathcal{T}_a$ captured (K_{ab}, $K_{\mathcal{M}}$ and its credential known)		
Q17–Q18	first design: as Q19, and GK_b secret	✗ attack
Q19–Q20	acceptance ⇒ member started it	✓ true
Q21	member completes ⇒ $\mathcal{T}_b$ sent that reply	✓ true*
Q22	secrecy of GK_b	✓ true
Q23	member's data after its long-term key leaks	✓ true*

* false without the signed reply (A7)

Q17–Q18 hand the attacker K_{ab} and $K_{\mathcal{M}}$: with tickets alone ProVerif finds A2. Q19–Q23 add the member's signature, $\mathcal{T}_a$'s own credential and the fix for A7, and all hold. Without that fix, Q21 and Q23 fail: this is how A7 was found.

VII. IMPLEMENTATION AND EVALUATION

Setup. Both schemes run on BN254 with `py_ecc` (pure Python) and `cryptography`, in one process on one laptop. The base scheme's symmetric pairing is realised by placing the argument the TA can compute in $\mathbb{G}_2$. For each n the two schemes run back to back. Because the laptop's speed drifted during the day, we also report a load-independent ratio from the exact operation counts recorded by the code. Here a pairing costs about 44 scalar multiplications, against about 8 with MIRACL. BN254 gives roughly 100-bit security today, so a deployment would use a 128-bit curve; we use it for one curve across both schemes. Fifty-seven tests pass, covering correctness and every attack on both designs. E1–E6 measure the first design. E8 runs both designs side by side

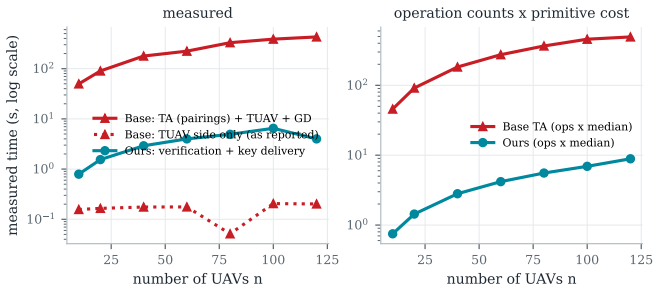


Fig. 1. E2: time to authenticate and key n UAVs (log scale). Left: measured. Right: operation counts × median costs.

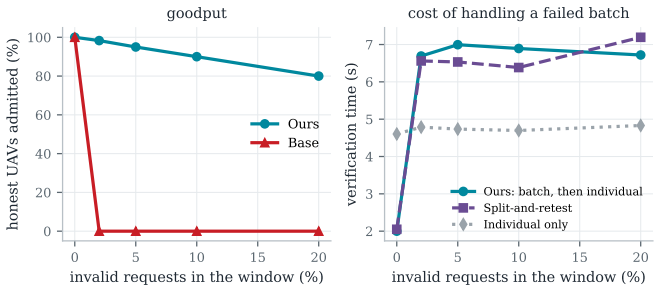


Fig. 2. E3: honest UAVs admitted and verification time under batch poisoning ($n = 60$).

and reports the cost of the fixes, and its absolute times reflect a different machine state from E1–E6, so we compare within each experiment only.

Latency (E2). For $n = 100$ our TUAV authenticates and keys the swarm in 6.49 s; the base scheme needs 388 s before any satellite delay, and its TUAV-only part (the part [1] reports) is 0.05% of that. By operation count the base scheme does 55–66× more work at every n; measured ratios ($n \leq 100$) are 56–67×. With the base paper's MIRACL constants the figures are 228 ms against 3720 ms plus RTT, about 16× (Fig. 1).

Where the work falls. Total work is not the drone's work. At $n = 100$ the base scheme's TUAV spends about 3.25 s (GA3 and GD, from single runs), roughly half of our 6.49 s. We do not lighten the TUAV. We remove the TA's pairings and the satellite trip, and move verification onto a device that the tether powers.

Availability (E1). The base scheme succeeds in a fraction A of rounds; with three retries at $A = 50\%$ it reaches 87.5% under independent losses but only 59.2% under bursty outages. Ours does not use the link.

Poisoning (E3). One invalid request blocks every honest UAV in the base scheme. Ours admits every UAV whose own request is intact, at every attack rate (E8 covers forged requests that name an honest pseudonym), with the cost bounded by a failed batch plus individual checks (2.00 s clean, 7.00 s worst). Split-and-retest [5] brings no gain because pairing-free batching is only a few times cheaper per signature (Fig. 2).

Traffic (E4). For $n = 120$ the base scheme's acknowledgements carry 17.5 MB of CRT coefficients and it moves

158.6 KB over the satellite link. Our single ACK is 9.6 KB (1,823× less). Our join request is larger (225 B against 137 B).

Vehicles and handover (E5, E6). A bundle of 500 vehicles is admitted, including keys, in 29.6 s, and non-member bundles are dropped in microseconds. A ticket handover costs the new TUAV 52.3 ms instead of 105.0 ms. Over a one-hour mobility run with 557 cell crossings, the mean service gap falls from 756 ms to 103 ms. Of that gap, 500 ms is the assumed mean wait for the next beacon (1 s period), not a measurement.

Attacks (E8a). All 7 attack variants succeed on the first design and 0 of 7 succeed on the hardened design.

Lockout (E8b). On the first design, one forged request locks a victim out for the rest of the epoch. On the hardened design it costs nothing lasting. With 4 forged requests per window the victim is admitted in the first window 39.0% of the time (predicted 40%) and within five windows 94.5% of the time; with 32 it still gets in within five windows 29.5% of the time (predicted 26.8%; the standard error is at most 3.5 points over 200 trials). Holding a victim below 50% over five windows takes about 15 forged requests in every window, instead of one packet in a 15-minute epoch. That is close to jamming, which no protocol prevents. When every pseudonym in a swarm of 60 is spoofed once, the first design admits 0 and evicts 60; the hardened design admits 60 and evicts none, at 1.43 s against 0.36 s for a clean window (3.9×), as Proposition 2 bounds.

Blast radius (E8c). On a 25-cell grid, one captured TUAV exposes 15.7% of cells on the first design and 4.0% (its own) on the hardened design; 3 captured TUAVs expose 43.2% and 12.0% (Fig. 3). With 3 captured, the first design also lets them displace 7.8 members and frame 7.8 on average, one attempted per neighbouring cell for each attack; the hardened design allows 0 and 0.

Cost of the fixes (E8d). Containment is not free, and it removes C3's main computational saving. In a 10-member cell the new TUAV spends 138.11 ms on a signature-bound ticket, against 44.11 ms for a ticket alone and 95.31 ms for full re-admission, because it verifies a signature, does a fresh ECDH and rekeys the cell. The first design's saving at the new TUAV was the price of trusting the neighbour. The member, which now also checks the new TUAV's signed reply (A7), spends 138.01 ms against 138.36 ms, so its saving is gone too. The computation in one handover is 276 ms against 234 ms. What remains is latency: full re-admission must first wait for a beacon, 734 ms with the assumed 1 s period. These handover figures were re-measured after the fix on a slower machine than the rest of E8; compare them with each other only. Each crossing also adds 65 B for the member's signature, 131 B for the reply's signature and public key and 800 B per 10-member cell rekeyed, and the old cell's rekey takes 0.23 ms. With half of 60 UAVs arriving, key delivery takes 282.6 ms against 274.8 ms, and sends 4800 B against 2400 B. An eviction certificate is 310 B and 101.97 ms to check at $t = 2$, and 775 B and 189.01 ms at $t = 5$.

VIII. Limitations

A malicious TA keeps Girault level-2 power (detectable key replacement and phantom enrolment). A captured TUAV still exposes its own cell's keys and can run a rogue cell, until it is evicted. C4 assumes that someone detects the capture, that t honest TUAVs can reach each other, and that fewer than t are captured; t or more captured TUAVs can evict honest ones. We provide no way to restore an evicted TUAV without the TA. An outsider can still flood a window with fresh pseudonyms, and a TUAV needs a verification budget against this, which we do not evaluate. Members arriving within one window share that window's fresh key. Pseudonyms are linkable within an epoch. Requests are larger. Our timings come from a pure-Python library on a laptop whose speed drifted, network effects are simulated, and the security arguments are sketches with symbolic models. The ProVerif models cover the signature-bound handover with a captured TUAV but not C4. An embedded constant-time implementation, a computational proof of containment, and manufacturer-bound keys (level 3) are future work.

IX. Conclusion

The TUAV architecture of [1] is a good answer to infrastructure loss, but its protocol keeps the cloud authority inside every step. Moving the trust root offline restores infrastructure independence, escrow-freeness, forward secrecy and secure revocation together, reduces total work and traffic, and extends naturally to vehicles and to multi-TUAV handover. It also removes the party that could overrule a forgery or disown a stolen device, and our own first design shows what follows. Requiring checkable evidence for every lasting state change contains a captured TUAV to its own cell, at the measured cost of the handover's CPU saving; its latency saving remains.

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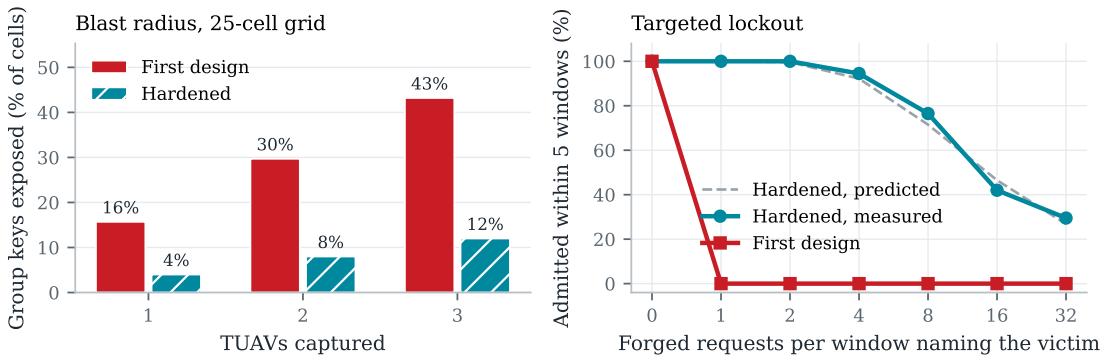


Fig. 3. E8. Left: cells whose group key one to 3 captured TUAVs can read (25-cell grid, mean of 30 random placements, attacks executed). Right: chance that a victim is admitted within five 1 s windows while forged requests name it, over 200 trials. The first design's line assumes one forged request arrives ahead of the victim's.

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